Data Communication & Computer Networks (CN)

Topic 5 of 7 : Transmission Medium

43 MCQs — Repo Exam Questions + Forouzan — bilingual (English + Hindi), exam-ready

Reference Book: **Data Communications and Networking — Behrouz A. Forouzan**

Prepared for: **RPSC Programmer • ACP • RSMSSB IA • SA • DD • BCI • SCI • REET • RAS • DSSSB • KVS • NVS • HTET • SSC CGL/CHSL • RRB • IBPS • UPSC & all Rajasthan + Central Govt. Exams (Computer Awareness)**

Total Questions: **43** MCQs + **23** Nukte rapid-fire revision points

Question Type: MCQ — 4 options (A, B, C, D) per question, in English + Hindi

Layout: Left = **question** | Right = **explanation** (green): ✔ **why the correct answer is right** + ★ **Nukta** (key exam point)

Language rule: Hindi is the second language; **technical terms stay in English** (topology, protocol, jitter…)

Verification: Every answer cross-verified **4×** (Forouzan + official answer keys of Subject Test / Unit Test Solutions). Wrong options verified internally.

Syllabus Coverage — sub-topic wise order

1. Section A — Signals & Performance (Encoding, Noise, Bandwidth, Data Rate) — Q1 - Q20

2. Section B — Guided Media (UTP, Coaxial, Fiber Optic) — Q21 - Q36

3. Section C — Unguided Media (Wireless) — Q37 - Q43

How to read: Left side: question + 4 options (try it yourself first). Right side (green): ✔ correct answer with why it is right, and ★ **Nukta** — the one-line exam tip to remember. Practice in 2 passes: first read question + options only, then check the green column.

System: generated by **tools/cn_build_topic.py** from **tools/data/topic5_data.py**. To add more questions later: drop PDFs into **Computer Network/Incoming/** and run **tools/cn_pipeline.py** — new questions auto-classify topic-wise. See **Computer Network/README.md**.

Section A — Signals & Performance (Encoding, Noise, Bandwidth, Data Rate)

Line coding (Manchester, 8B/10B, scramblers), impairments (attenuation/crosstalk/noise), signal parameters, data-rate & Shannon/Nyquist calculations.

Q1. 'Manchester Encoding' and 'Differential Manchester Encoding' are signal encoding techniques used at which layer?

'Manchester Encoding' और 'Differential Manchester Encoding' signal encoding techniques किस layer पर उपयोग होती हैं?

(A) Physical Layer **(B)** Data Link Layer
(C) Network Layer **(D)** Presentation Layer

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)]

✔ **Correct: (A) Physical Layer** — Line coding (Manchester, differential Manchester, NRZ, AMI) converts bits into signals — a Physical layer job.

Line coding (Manchester, differential Manchester, NRZ, AMI) bits को signals में बदलती है — यह Physical layer का काम है।

★ **Nukta:** Physical layer = signalling & encoding (Manchester = mid-bit transition, self-clocking).

Physical layer = signalling और encoding (Manchester = mid-bit transition, self-clocking)।

Q2. A channel has high bandwidth but suffers from high noise. Which encoding scheme gives better error resistance?

एक channel की bandwidth अधिक है लेकिन noise बहुत है। कौन-सी encoding scheme बेहतर error resistance देगी?

(A) NRZ **(B)** Manchester
(C) AMI **(D)** ASK

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)]

✔ **Correct: (B) Manchester** — Manchester encoding has a transition in every bit period — the receiver resynchronises quickly and detects errors easily, giving strong noise resilience.

Manchester encoding में हर bit period में transition होता है — receiver जल्दी resynchronise होता है और errors आसानी से पकड़ लेता है, इसलिए noise resilience मज़बूत है।

★ **Nukta:** Manchester = self-clocking + DC-free + strong error visibility. NRZ = long runs → clock loss in noise.

Manchester = self-clocking + DC-free + मज़बूत error visibility। NRZ = लंबे runs → noise में clock loss।

Q3. Which of the following Ethernet encodings has the highest redundancy?

निम्न में से कौन-सी Ethernet encoding की redundancy सबसे अधिक है?

(A) 1B2B **(B)** 4B5B
(C) 5B6B **(D)** 8B10B

[Source: Unit Test-4 Solution (OCR), p9 (options recovered from source)]

✔ **Correct: (A) 1B2B** — 1B2B encodes every 1 bit as 2 bits — 100% redundancy, the highest of the four (4B5B = 25%, 5B6B = 20%, 8B10B = 25%).

1B2B हर 1 bit को 2 bits के रूप में encode करती है — 100% redundancy, चारों में सबसे अधिक (4B5B = 25%, 5B6B = 20%, 8B10B = 25%)।

★ **Nukta:** Redundancy (overhead): 1B2B 100% > 4B5B/8B10B 25% > 5B6B 20%. More redundancy = better clocking/error properties, more bandwidth.

Redundancy (overhead): 1B2B 100% > 4B5B/8B10B 25% > 5B6B 20%। ज़्यादा redundancy = बेहतर clocking/error properties, ज़्यादा bandwidth।

Q4. Which encoding technique is used to transmit signals in Gigabit Ethernet over fiber optic medium?

Fiber optic medium पर Gigabit Ethernet में signal भेजने के लिए कौन-सी encoding technique उपयोग होती है?

(A) Differential Manchester encoding **(B)** Non-Return-to-Zero (NRZ)

(C) 4B/5B **(D)** 8B/10B

[Source: Unit Test-4 Solution (OCR), p30 (options recovered from source)]

✔ **Correct: (D) 8B/10B** — Gigabit Ethernet (1000BASE-X over fiber) uses 8B/10B encoding — 8 data bits are carried as 10 code bits for clocking and DC balance.

Gigabit Ethernet (fiber पर 1000BASE-X) 8B/10B encoding उपयोग करता है — 8 data bits को 10 code bits के रूप में भेजा जाता है (clocking और DC balance के लिए)।

★ **Nukta:** 100BASE-TX uses 4B/5B + MLT-3; 1000BASE-X uses 8B/10B; 10GBASE-R uses 64B/66B. Ye exam favourite ladder hai.

100BASE-TX में 4B/5B + MLT-3; 1000BASE-X में 8B/10B; 10GBASE-R में 64B/66B। यह exam favourite ladder है।

Q5. The primary use of scramblers in line coding schemes is to:

Line coding schemes में scramblers का मुख्य उपयोग क्या है?

(A) Make the data look random **(B)** Compress transmission data

(C) Increase signal bandwidth **(D)** Decrease signal bandwidth

[Source: RPSC Programmer 2024 Paper-1 (official key)]

✔ **Correct: (A) Make the data look random** — A scrambler randomises the transmitted bit pattern (breaks long runs of 0s/1s) so the receiver recovers the clock easily — it does NOT compress or change bandwidth.

Scrambler transmitted bit pattern को randomise करता है (लंबे 0s/1s के runs तोड़ता है) ताकि receiver आसानी से clock recover कर सके — यह compress या bandwidth नहीं बदलता।

★ **Nukta:** Official RPSC 2024 key. Scrambler = random-looking line signal (clock recovery). Not compression, not encryption.

Official RPSC 2024 key। Scrambler = random-looking line signal (clock recovery)। Compression नहीं, encryption नहीं।

Q6. Reduction in signal strength over distance is called:

दूरी के साथ signal strength में कमी को क्या कहते हैं?

(A) Distortion **(B)** Noise

(C) Attenuation **(D)** Interference

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch) · also CN Topic Test-04]

✔ **Correct: (C) Attenuation** — Attenuation is the gradual loss of signal strength (amplitude) as it travels along a medium — measured in dB/km (or dB/100 ft).

Attenuation medium के साथ यात्रा करते समय signal strength (amplitude) का क्रमिक नुकसान है — इसे dB/km (या dB/100 ft) में मापा जाता है।

★ **Nukta:** Impairments: Attenuation (strength loss), Distortion (shape change), Noise (unwanted signals). Attenuation ↑ with distance & frequency.

Impairments: Attenuation (strength loss), Distortion (shape change), Noise (unwanted signals)। Attenuation ↑ distance और frequency के साथ।

Q7. In a copper cable, attenuation increases with frequency. Which change will REDUCE attenuation?

Copper cable में attenuation frequency के साथ बढ़ता है। कौन-सा बदलाव attenuation घटाएगा?

(A) Increase cable length **(B)** Use lower-category UTP

(C) Use thicker copper core **(D)** Increase signal frequency

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)]

✔ **Correct: (C) Use thicker copper core** — A thicker copper core has lower electrical resistance — so the signal loses less strength over distance.

मोटे copper core का electrical resistance कम होता है — इसलिए signal को दूरी पर कम strength गँवानी पड़ती है।

★ **Nukta:** Reduce attenuation: shorter cable, thicker/better copper, higher category cable, lower frequency, or add repeaters/amplifiers.

Attenuation घटाना: छोटी cable, मोटा/बेहतर copper, उच्च श्रेणी cable, कम frequency, या repeaters/amplifiers जोड़ें।

Q8. Which noise is caused by electromagnetic coupling between nearby cables?

पास की cables के बीच electromagnetic coupling से कौन-सा noise होता है?

(A) Thermal noise **(B)** Crosstalk

(C) Impulse noise **(D)** Quantization noise

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)]

✔ **Correct: (B) Crosstalk** — Crosstalk is the unwanted signal leakage from one wire pair to a neighbouring pair caused by electromagnetic coupling.

Crosstalk एक wire pair से पास की दूसरी pair में electromagnetic coupling के कारण होने वाला unwanted signal leakage है।

★ **Nukta:** Noise types: Thermal (random electron motion), Crosstalk (pair-to-pair coupling), Impulse (sudden spikes), Quantization (PCM sampling).

Noise types: Thermal (random electron motion), Crosstalk (pair-to-pair coupling), Impulse (sudden spikes), Quantization (PCM sampling)।

Q9. Which impairment cannot be completely eliminated but only minimised?

कौन-सा impairment पूरी तरह समाप्त नहीं किया जा सकता, केवल कम किया जा सकता है?

(A) Attenuation **(B)** Distortion

(C) Noise **(D)** Delay

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)]

✔ **Correct: (C) Noise** — Noise can never be fully eliminated — it can only be minimised (shielding, better cable, lower power, error control).

Noise कभी पूरी तरह समाप्त नहीं हो सकती — केवल कम की जा सकती है (shielding, बेहतर cable, कम power, error control)।

★ **Nukta:** Attenuation/distortion can be fixed with repeaters/equalisers; noise is fundamental (thermal floor) — only minimised.

Attenuation/distortion repeaters/equalisers से ठीक हो सकते हैं; noise मूलभूत है (thermal floor) — केवल कम होती है।

Q10. Which parameter of an analog signal represents the number of cycles per second?

Analog signal का कौन-सा parameter प्रति सेकंड cycles की संख्या दर्शाता है?

(A) Amplitude **(B)** Phase
(C) Frequency **(D)** Wavelength

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)]

✔ **Correct: (C) Frequency** — Frequency is the number of signal cycles completed per second, measured in Hertz (Hz).

Frequency प्रति सेकंड पूरे होने वाले signal cycles की संख्या है, जिसे Hertz (Hz) में मापा जाता है।

★ **Nukta:** Signal parameters: Amplitude (height), Frequency (cycles/sec), Phase (start position), Wavelength (distance/cycle = speed/frequency).

Signal parameters: Amplitude (ऊँचाई), Frequency (cycles/sec), Phase (शुरुआती स्थिति), Wavelength (दूरी/cycle = speed/frequency)।

Q11. The period of a signal is 100 ms. What is its frequency in kilohertz?

एक signal की period 100 ms है। इसकी frequency kilohertz में कितनी होगी?

(A) 10^{-1} kHz **(B)** 10^{-2} kHz
(C) 10^{-3} kHz **(D)** 10^{-4} kHz

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)]

✔ **Correct: (B) 10^{-2} kHz** — Frequency f = 1/T = 1/0.1 s = 10 Hz = 0.01 kHz = 10^{-2} kHz.

Frequency f = 1/T = 1/0.1 s = 10 Hz = 0.01 kHz = 10^{-2} kHz।

★ **Nukta:** f = 1/T. 100 ms = 0.1 s → 10 Hz. Convert Hz→kHz by dividing by 1000.

f = 1/T। 100 ms = 0.1 s → 10 Hz। Hz→kHz के लिए 1000 से भाग दें।

Q12. _____ signals are discrete and are generated by digital modulation.

_____ signals discrete होते हैं और digital modulation से बनते हैं।

(A) Analog **(B)** Digital
(C) Noise **(D)** Bandwidth

[Source: Random Test- Rajasthan PYQ (OCR), p5 — Q108 (options recovered)]

✔ **Correct: (B) Digital** — Digital signals are discrete (defined only at instants) and are produced by digital modulation techniques.

Digital signals discrete होते हैं (केवल कुछ क्षणों पर परिभाषित) और digital modulation techniques से बनते हैं।

★ **Nukta:** Rajasthan PYQ verified key (B). Analog = continuous; Digital = discrete. Ye sabse basic signal distinction hai.

Rajasthan PYQ verified key (B)। Analog = continuous; Digital = discrete। यह signal का सबसे बुनियादी भेद है।

Q13. Quantizing noise occurs in:

Quantizing noise किसमें होती है?

(A) PCM (Pulse Code Modulation) **(B)** TDM
(C) FDM **(D)** PPM

[Source: Random Test- Rajasthan PYQ (OCR), p10 — Q39 (options recovered)]

✔ **Correct: (A) PCM (Pulse Code Modulation)** — Quantizing noise is the small error introduced when an analog sample is rounded to the nearest discrete level in PCM.

Quantizing noise वह छोटी त्रुटि है जो PCM में analog sample को निकटतम discrete level पर गोल करने से आती है।

★ **Nukta:** Rajasthan PYQ verified key (A). PCM = sampling + quantizing + encoding → quantizing error = quantization noise.

Rajasthan PYQ verified key (A)। PCM = sampling + quantizing + encoding → quantizing error = quantization noise।

Q14. If the channel bandwidth is limited to 6 kHz and the signal-to-noise ratio is 16, what is the channel capacity?

यदि channel bandwidth 6 kHz सीमित है और signal-to-noise ratio 16 है, तो channel capacity कितनी होगी?

(A) 15.15 kbps **(B)** 24.74 kbps

(C) 30.12 kbps **(D)** 52.18 kbps

[Source: Random Test- Rajasthan PYQ (OCR), p30 — Q45 (options recovered)]

✔ **Correct: (B) 24.74 kbps** — By Shannon's capacity formula $C = B \times \log_2(1 + SNR) = 6000 \times \log_2(17) \approx 6000 \times 4.09 \approx 24.5\text{-}24.7$ kbps — matching 24.74 kbps.

Shannon capacity सूत्र से $C = B \times \log_2(1 + SNR) = 6000 \times \log_2(17) \approx 6000 \times 4.09 \approx 24.5\text{-}24.7$ kbps — 24.74 kbps से मेल खाता है।

★ **Nukta:** Rajasthan PYQ verified key (B). Shannon: $C = B \log_2(1+SNR)$. Nyquist: $C = 2B \log_2(L)$ (noiseless).

Rajasthan PYQ verified key (B)। Shannon: $C = B \log_2(1+SNR)$। Nyquist: $C = 2B \log_2(L)$ (noiseless)।

Q15. Given a bit rate of b bits/sec, the time required to send 16 bits is:

यदि bit rate b bits/sec है, तो 16 bits भेजने में कितना समय लगेगा?

(A) 16 × b sec **(B)** 16/b sec

(C) 16b sec **(D)** b/16 sec

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)]

✔ **Correct: (B) 16/b sec** — Time = number of bits ÷ bit rate = 16 ÷ b = 16/b seconds.

Time = bits की संख्या ÷ bit rate = 16 ÷ b = 16/b seconds।

★ **Nukta:** Time = size/rate. 1000 bits at 100 bps = 10 s. Units: bits ÷ (bits/sec) = sec.

Time = size/rate। 1000 bits at 100 bps = 10 s। Units: bits ÷ (bits/sec) = sec।

Q16. An analog signal carries 4 bits per signal unit. If 1000 signal units are sent per second, what are the baud rate and bit rate?

एक analog signal प्रति signal unit 4 bits ले जाती है। यदि प्रति सेकंड 1000 signal units भेजे जाएँ, तो baud rate और bit rate क्या होंगे?

(A) 1000 bauds/sec, 4000 bps **(B)** 1000 bauds/sec, 500 bps

(C) 4000 bauds/sec, 1000 bps **(D)** 2000 bauds/sec, 1000 bps

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)]

✔ **Correct: (A) 1000 bauds/sec, 4000 bps** — Baud rate = signal units per second = 1000 bauds/sec. Bit rate = baud × bits per unit = 1000 × 4 = 4000 bps.

Baud rate = प्रति सेकंड signal units = 1000 bauds/sec। Bit rate = baud × प्रति unit bits = 1000 × 4 = 4000 bps।

★ **Nukta:** Bit rate = Baud rate × bits per signal unit. Baud = signalling speed; bps = data speed.

Bit rate = Baud rate × प्रति signal unit bits। Baud = signalling speed; bps = data speed।

Q17. The effective bandwidth of a communication channel is primarily limited by:

Communication channel की effective bandwidth मुख्य रूप से किससे सीमित होती है?

(A) Protocol overhead **(B)** Noise and attenuation

(C) File size **(D)** MAC address filtering

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)]

✔ **Correct: (B) Noise and attenuation** — Noise (with attenuation) limits how much data can be reliably pushed through a channel — Shannon's law shows capacity grows with bandwidth and SNR but is capped by noise.

Noise (attenuation के साथ) तय करती है कि channel से कितना data भरोसेमंद तरीके से भेजा जा सकता है — Shannon का नियम दिखाता है कि capacity bandwidth और SNR के साथ बढ़ती है पर noise से सीमित रहती है।

★ **Nukta:** Effective bandwidth is a physics/SNR limit. Protocol overhead reduces goodput but is not the channel's fundamental limit.

Effective bandwidth एक physics/SNR सीमा है। Protocol overhead goodput घटाता है पर channel की मूल सीमा नहीं।

Q18. What technique allows multiple signals to share the same communication channel?

कौन-सी technique कई signals को एक ही communication channel share करने देती है?

(A) Multiplexing **(B)** Phase modulation

(C) Analog modulation **(D)** Digital modulation

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (A) Multiplexing** — Multiplexing combines multiple signals onto one shared medium (FDM/TDM/WDM/CDM) — a demultiplexer separates them at the receiving end.

Multiplexing कई signals को एक shared medium पर जोड़ती है (FDM/TDM/WDM/CDM) — receiving end पर demultiplexer उन्हें अलग करता है।

★ **Nukta:** Multiplexing = many share one channel. FDM = frequency bands, TDM = time slots, WDM = light wavelengths, CDM = codes.

Multiplexing = कई एक channel share करें। FDM = frequency bands, TDM = time slots, WDM = light wavelengths, CDM = codes।

Q19. Communication using a modulated carrier to apply a data signal to a transmission medium in analog form is:

Modulated carrier का उपयोग करके data signal को analog रूप में transmission medium पर डालना क्या कहलाता है?

(A) Broadband networking **(B)** Baseband networking

(C) Modem **(D)** Local area network

[Source: IBPS RRB PO Computer Awareness PYQ (via Meritnotes)]

✔ **Correct: (A) Broadband networking** — Broadband networking sends data as a modulated analog carrier; baseband sends digital signals directly on the medium.

Broadband networking data को modulated analog carrier के रूप में भेजता है; baseband digital signals सीधे medium पर भेजता है।

★ **Nukta:** Baseband = digital signalling (Manchester etc.). Broadband = analog carriers + modulation (FDM channels).

Baseband = digital signalling (Manchester आदि)। Broadband = analog carriers + modulation (FDM channels)।

Q20. The amount of data transmitted in a fixed amount of time is called:

निश्चित समय में भेजे गए data की मात्रा को क्या कहते हैं?

(A) Refresh rate **(B)** Resolution
(C) Bandwidth **(D)** Frequency

[Source: Unit Test-4 Solution (OCR), p23]

✔ **Correct: (C) Bandwidth** — Bandwidth (in this exam phrasing) is the data transmitted per unit time — the communication channel's capacity (bps).

इस exam शैली में Bandwidth प्रति इकाई समय भेजा गया data है — communication channel की capacity (bps)।

★ **Nukta:** Strictly: THROUGHPUT = data actually sent per second; BANDWIDTH = maximum capacity. Banks often call throughput 'bandwidth' — read options carefully.

कठोरता से: THROUGHPUT = वास्तव में प्रति सेकंड भेजा data; BANDWIDTH = अधिकतम capacity। Banks अक्सर throughput को 'bandwidth' कहते हैं — options ध्यान से पढ़ें।

Section B — Guided Media (UTP, Coaxial, Fiber Optic)

Twisted pair, coaxial and fiber — properties, connectors, distances, termination, EMI behaviour, total internal reflection.

Q21. Assertion (A): Fiber optic cable is immune to electromagnetic interference (EMI). Reason (R): It transmits data in the form of light pulses rather than electrical signals. Which option is correct?

Assertion (A): Fiber optic cable electromagnetic interference (EMI) से मुक्त है। Reason (R): यह data electrical signals के बजाय light pulses के रूप में भेजती है। कौन-सा विकल्प सही है?

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A) **(B)** Both (A) and (R) are true, but (R) is NOT the correct explanation
(C) (A) is true, but (R) is false **(D)** (A) is false, but (R) is true

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch) — Assertion-Reason]

✔ **Correct: (A) Both (A) and (R) are true, and (R) is the correct explanation of (A)** — Fiber is immune to EMI (A is true) precisely BECAUSE it carries light, not electrical current (R is true and explains A).

Fiber EMI से मुक्त है (A सही) क्योंकि यह electrical current नहीं, प्रकाश ले जाता है (R सही और A की व्याख्या करता है)।

★ **Nukta:** Fiber = light pulses → EMI-proof + tap-proof + huge bandwidth. Copper = electrical → EMI/crosstalk prone.

Fiber = light pulses → EMI-proof + tap-proof + विशाल bandwidth। Copper = electrical → EMI/crosstalk प्रवण।

Q22. Which statements are TRUE for fiber optic cables? I. Immune to tapping II. High bandwidth III. Requires electrical grounding IV. Works using total internal reflection

Fiber optic cables के लिए कौन-से कथन सही हैं? I. Tapping से मुक्त II. उच्च bandwidth III. Electrical grounding चाहिए IV. Total internal reflection से काम करती है

(A) I, II and IV **(B)** II and III

(C) I, III and IV **(D)** I and II

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch) — Q33]

✔ **Correct: (A) I, II and IV** — Fiber carries light (hard to tap, huge bandwidth, works on total internal reflection) and being glass/dielectric it needs NO electrical grounding.

Fiber प्रकाश ले जाती है (tapping कठिन, विशाल bandwidth, total internal reflection पर काम) और काँच/dielectric होने से इसे electrical grounding की आवश्यकता नहीं।

★ **Nukta:** Fiber truths: immune to tapping & EMI, highest bandwidth, TIR propagation. Myths: grounding (NO), affected by lightning (NO).

Fiber truths: tapping और EMI से मुक्त, सर्वोच्च bandwidth, TIR propagation। Myths: grounding (नहीं), lightning (नहीं)।

Q23. Optical fiber uses reflection to guide light through a channel. At total internal reflection, the angle of incidence is _____ the critical angle.

Optical fiber प्रकाश को channel में भेजने के लिए reflection उपयोग करती है। Total internal reflection में angle of incidence, critical angle से _____ होता है।

(A) Equal to **(B)** Less than

(C) Greater than **(D)** Less than or equal to

[Source: Repo: CN Topic Test-04 (UNIT TEST-4) — recovered from BCI T8 Q33 options]

✔ **Correct: (C) Greater than** — Total internal reflection happens only when the angle of incidence is GREATER than the critical angle — then light stays inside the core.

Total internal reflection केवल तब होती है जब angle of incidence critical angle से अधिक हो — तब प्रकाश core के भीतर ही रहता है।

★ **Nukta:** TIR rule: angle of incidence > critical angle (denser → rarer medium). Ye fiber propagation ka base hai.

TIR rule: angle of incidence > critical angle (denser → rarer medium)। यह fiber propagation का आधार है।

Q24. A company needs long-distance communication (5-10 km), high data rate and immunity from electromagnetic interference. Which medium is best?

एक कंपनी को लंबी दूरी (5-10 km) संचार, उच्च data rate और EMI से मुक्ति चाहिए। कौन-सा medium सर्वोत्तम है?

(A) UTP **(B)** Coaxial cable

(C) Single-mode fiber **(D)** Multi-mode fiber

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)]

✔ **Correct: (C) Single-mode fiber** — Single-mode fiber carries one light ray tens of kilometres with very high data rates and is completely immune to EMI.

Single-mode fiber एक light ray को दसियों किलोमीटर तक बहुत उच्च data rates के साथ ले जाती है और पूरी तरह EMI से मुक्त है।

★ **Nukta:** Fiber choices: Multi-mode = short (LANs, few hundred m). Single-mode = long haul (km) + highest rate. Copper = short & EMI-prone.

Fiber विकल्प: Multi-mode = छोटी दूरी (LANs, कुछ सौ m)। Single-mode = लंबी (km) + उच्चतम rate। Copper = छोटी और EMI-प्रवण।

Q25. A network uses coaxial cable but suffers from signal reflection. What is the most likely cause?

एक network coaxial cable उपयोग करता है लेकिन signal reflection से पीड़ित है। सबसे संभावित कारण क्या है?

(A) EMI **(B)** Incorrect termination resistance

(C) Crosstalk **(D)** Low bandwidth

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)]

✔ **Correct: (B) Incorrect termination resistance** — Coaxial cable must be terminated with its characteristic impedance (e.g., 50Ω/75Ω); a mismatch causes signal reflections back down the cable.

Coaxial cable को उसकी characteristic impedance (जैसे 50Ω/75Ω) से terminate करना अनिवार्य है; mismatch होने पर signal cable में वापस reflect होते हैं।

★ **Nukta:** Coax rule: proper termination (BNC/terminator) prevents reflections. Un-terminated bus = standing waves & echoes.

Coax rule: सही termination (BNC/terminator) reflections रोकता है। Un-terminated bus = standing waves और echoes।

Q26. A UTP cable used in Gigabit Ethernet shows high delay skew. What is the cause?

Gigabit Ethernet में उपयोग UTP cable में delay skew अधिक है। कारण क्या है?

(A) Different twist rates in the cable pairs **(B)** Excessive EMI

(C) Fiber splice error **(D)** Wrong wireless channel

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)]

✔ **Correct: (A) Different twist rates in the cable pairs** — Each pair in a UTP cable is twisted at a slightly different rate (to reduce crosstalk) — so signals arrive with small timing differences, called delay skew.

UTP cable में हर pair थोड़ी अलग दर से twisted होता है (crosstalk घटाने के लिए) — इसलिए signals थोड़े अलग समय पर पहुँचते हैं, इसे delay skew कहते हैं।

★ **Nukta:** Delay skew = pair-to-pair propagation time difference (tight twist → shorter wire → less skew). Cat6 has tighter specs for Gigabit.

Delay skew = pair-to-pair propagation time अंतर (tight twist → छोटी wire → कम skew)। Cat6 में Gigabit के लिए कड़े specs हैं।

Q27. Which of the following best explains why fiber optic cables have a lower bit error rate than copper cables?

Fiber optic cables की copper cables की तुलना में कम bit error rate की सबसे अच्छी व्याख्या कौन-सी है?

(A) Higher signal power **(B)** Immunity to electromagnetic interference

(C) Simpler modulation techniques **(D)** Larger physical size

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)]

✔ **Correct: (B) Immunity to electromagnetic interference** — Fiber carries light and is immune to EMI/crosstalk/impulse noise — the main causes of bit errors in copper — so its BER is far lower.

Fiber प्रकाश ले जाती है और EMI/crosstalk/impulse noise से मुक्त है — जो copper में bit errors के मुख्य कारण हैं — इसलिए इसका BER बहुत कम होता है।

★ **Nukta:** Fiber wins on: EMI immunity, low attenuation, huge bandwidth, tap security.

Fiber में श्रेष्ठता: EMI immunity, कम attenuation, विशाल bandwidth, tap security।

Q28. Which transmission medium works on the principle of total internal reflection?

कौन-सा transmission medium total internal reflection के सिद्धांत पर काम करता है?

(A) Optical fiber cable **(B)** Shielded twisted pair

(C) Unshielded twisted pair **(D)** Coaxial cable

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)]

✔ **Correct: (A) Optical fiber cable** — In optical fiber, light stays inside the glass core by total internal reflection (angle of incidence > critical angle).

Optical fiber में प्रकाश total internal reflection के कारण glass core के भीतर रहता है (angle of incidence > critical angle)।

★ **Nukta:** Fiber = TIR. Copper media use electrical signals (electromagnetic fields), not light reflection.

Fiber = TIR। Copper media electrical signals (electromagnetic fields) उपयोग करते हैं, प्रकाश reflection नहीं।

Q29. Crosstalk in twisted pair cables is typically minimised by:

Twisted pair cables में crosstalk आमतौर पर किससे कम होता है?

(A) Increasing the power level **(B)** Increasing the twist rate

(C) Adding shielding to the core **(D)** Using longer cables

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)]

✔ **Correct: (B) Increasing the twist rate** — Tightening the twists equalises coupling and cancels interference between pairs — higher categories (Cat6/6a) use tighter twists.

Tight twists coupling को संतुलित करते हैं और pairs के बीच interference रद्द करते हैं — उच्च श्रेणियों (Cat6/6a) में tighter twists होते हैं।

★ **Nukta:** Anti-crosstalk tricks: tighter twists, pair separation, shielding (STP), lower power.

Anti-crosstalk tricks: tighter twists, pair separation, shielding (STP), कम power।

Q30. What is the typical maximum length of a UTP cable in Ethernet before signal degradation?

Signal degradation से पहले Ethernet में UTP cable की सामान्य अधिकतम लंबाई कितनी होती है?

(A) 50 meters **(B)** 100 meters

(C) 150 meters **(D)** 200 meters

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)]

✔ **Correct: (B) 100 meters** — Standard Ethernet UTP runs are limited to 100 metres (about 90 m of solid cable + 10 m of patch cords).

Standard Ethernet UTP runs 100 मीटर तक सीमित हैं (लगभग 90 m solid cable + 10 m patch cords)।

★ **Nukta:** Copper distance limits: UTP Ethernet 100 m, Thinnet 185 m, Thicknet 500 m, Fiber (multi-mode) ~550 m-2 km, (single-mode) tens of km.

Copper distance limits: UTP Ethernet 100 m, Thinnet 185 m, Thicknet 500 m, Fiber (multi-mode) ~550 m-2 km, (single-mode) दसियों km।

Q31. Which connector is commonly used for coaxial cable in physical networks?

Physical networks में coaxial cable के लिए सामान्यतः कौन-सा connector उपयोग होता है?

(A) RJ-45 **(B)** USB-C
(C) BNC **(D)** HDMI

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)]

✔ **Correct: (C) BNC** — Coaxial networks use BNC connectors (with a terminator at the ends); RJ-45 is for twisted pair, HDMI for displays, USB-C for peripherals.

Coaxial networks BNC connectors उपयोग करते हैं (छोर पर terminator के साथ); RJ-45 twisted pair के लिए, HDMI display के लिए, USB-C peripherals के लिए।

★ **Nukta:** Connectors: RJ-45 (UTP Ethernet), BNC (coax/Thinnet), SC/ST/LC (fiber), RJ-11 (telephone).

Connectors: RJ-45 (UTP Ethernet), BNC (coax/Thinnet), SC/ST/LC (fiber), RJ-11 (telephone)।

Q32. Which transmission medium is primarily used in modern internet backbone networks?

आधुनिक internet backbone networks में मुख्य रूप से कौन-सा transmission medium उपयोग होता है?

(A) Twisted pair cable **(B)** Coaxial cable
(C) Single mode fiber optic cable **(D)** Microwave links

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)]

✔ **Correct: (C) Single mode fiber optic cable** — Modern backbones use single-mode fiber — huge bandwidth over very long distances with low loss.

आधुनिक backbones single mode fiber उपयोग करते हैं — लंबी दूरी पर कम loss के साथ विशाल bandwidth।

★ **Nukta:** Backbone = single-mode fiber (undersea/terrestrial cables). LAN = UTP. Old cable TV = coax. Remote areas = microwave/satellite.

Backbone = single-mode fiber (undersea/terrestrial cables)। LAN = UTP। पुराना cable TV = coax। दूरस्थ क्षेत्र = microwave/satellite।

Q33. Which transmission medium is best for a network exposed to heavy electrical interference?

भारी electrical interference वाले network के लिए कौन-सा transmission medium सर्वोत्तम है?

(A) Unshielded twisted pair **(B)** Optical fiber
(C) Coaxial cable **(D)** Microwave

[Source: Repo: CN Topic Test-01 (UNIT TEST-1)]

✔ **Correct: (B) Optical fiber** — Optical fiber is immune to electrical/EMI interference because it transmits light, not electrical signals.

Optical fiber electrical/EMI interference से मुक्त है क्योंकि यह electrical signals नहीं, प्रकाश भेजती है।

★ **Nukta:** EMI-heavy areas (factories, power plants) → fiber. 'Immune to EMI' = keyword for fiber.

EMI-भारी क्षेत्र (कारखाने, power plants) → fiber। 'Immune to EMI' = fiber के लिए keyword।

Q34. Which of the following CANNOT be used as a medium for 802.3 Ethernet?

निम्न में से कौन-सा 802.3 Ethernet के लिए medium उपयोग नहीं किया जा सकता?

(A) Thin coaxial cable **(B)** Twisted pair cable
(C) Microwave link **(D)** Fiber optic cable

[Source: Repo: CN Topic Test-01 (UNIT TEST-1)]

✔ **Correct: (C) Microwave link** — 802.3 Ethernet is a wired standard — it runs on coax, twisted pair and fiber, but NOT on microwave (that is 802.11/ wireless).

802.3 Ethernet एक wired standard है — यह coax, twisted pair और fiber पर चलता है, लेकिन microwave पर नहीं (वह 802.11/wireless है)।

★ **Nukta:** 802.3 = Ethernet (wired: 10Base2/5, 10Base-T, 100Base-TX, 1000Base-X...). 802.11 = Wi-Fi (wireless).

802.3 = Ethernet (wired: 10Base2/5, 10Base-T, 100Base-TX, 1000Base-X...)। 802.11 = Wi-Fi (wireless)।

Q35. Which of the following is NOT a type of transmission medium?

निम्न में से कौन-सा transmission medium का प्रकार नहीं है?

(A) Coaxial cable **(B)** Fiber optics
(C) NIC **(D)** Twisted pair

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)]

✔ **Correct: (C) NIC** — A NIC (Network Interface Card) is a hardware adapter — a device, not a transmission medium.

NIC (Network Interface Card) एक hardware adapter है — एक device है, transmission medium नहीं।

★ **Nukta:** Media = guided (twisted pair, coax, fiber) + unguided (radio, microwave, satellite, infrared). NIC = adapter.

Media = guided (twisted pair, coax, fiber) + unguided (radio, microwave, satellite, infrared)। NIC = adapter।

Q36. Which cable has the maximum bandwidth?

किस cable की bandwidth अधिकतम होती है?

(A) Coaxial **(B)** Twisted pair
(C) Optical fiber **(D)** USB cable

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (C) Optical fiber** — Optical fiber offers the highest bandwidth — hundreds of Gbps to Tbps over long distances.

Optical fiber सर्वोच्च bandwidth देती है — लंबी दूरी पर सैकड़ों Gbps से Tbps तक।

★ **Nukta:** Bandwidth order: Fiber >> Coax > Twisted pair. USB is a peripheral bus, not a network cable type here.

Bandwidth क्रम: Fiber >> Coax > Twisted pair। USB peripheral bus है, यहाँ network cable type नहीं।

Section C — Unguided Media (Wireless)

Wi-Fi standards, infrared/microwave/Bluetooth, line-of-sight, CDMA sharing and its true/false statements.

Q37. Which of the following is a wireless networking technology?

निम्न में से कौन-सी wireless networking technology है?

(A) Ethernet **(B)** Wi-Fi
(C) Fiber Optic **(D)** DSL

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)]

✔ **Correct: (B) Wi-Fi** — Wi-Fi (IEEE 802.11) is a wireless LAN technology; Ethernet (802.3), fiber and DSL are all wired.

Wi-Fi (IEEE 802.11) एक wireless LAN technology है; Ethernet (802.3), fiber और DSL सब wired हैं।

★ **Nukta:** Wireless family: Wi-Fi (802.11), Bluetooth (802.15), WiMAX (802.16), mobile (GSM/LTE/5G).

Wireless family: Wi-Fi (802.11), Bluetooth (802.15), WiMAX (802.16), mobile (GSM/LTE/5G)।

Q38. Which wireless standard provides the highest data rate?

कौन-सा wireless standard सर्वोच्च data rate देता है?

(A) 802.11b **(B)** 802.11g
(C) 802.11n **(D)** 802.11ac

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)]

✔ **Correct: (D) 802.11ac** — 802.11ac (Wi-Fi 5) offers the highest rate among these — up to multi-Gbps (5 GHz, MU-MIMO, wide channels).

802.11ac (Wi-Fi 5) इनमें सर्वोच्च rate देता है — तक मल्टी-Gbps (5 GHz, MU-MIMO, wide channels)।

★ **Nukta:** Wi-Fi rates: b = 11 Mbps, g = 54 Mbps, n = 600 Mbps, ac = multi-Gbps. Order: b < g < n < ac.

Wi-Fi rates: b = 11 Mbps, g = 54 Mbps, n = 600 Mbps, ac = multi-Gbps। क्रम: b < g < n < ac।

Q39. Which type of wireless transmission requires line-of-sight?

किस प्रकार की wireless transmission के लिए line-of-sight आवश्यक है?

(A) Wi-Fi **(B)** Infrared
(C) Radio **(D)** Bluetooth

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)]

✔ **Correct: (B) Infrared** — Infrared transmission needs a direct line-of-sight (or reflection) between transmitter and receiver — it cannot pass through walls.

Infrared transmission के लिए transmitter और receiver के बीच सीधी line-of-sight (या reflection) चाहिए — यह दीवारों से नहीं गुजरती।

★ **Nukta:** IR = line-of-sight (TV remotes, IrDA). Radio/RF (Wi-Fi, Bluetooth) passes through walls (with loss).

IR = line-of-sight (TV remotes, IrDA)। Radio/RF (Wi-Fi, Bluetooth) दीवारों से गुजरते हैं (loss के साथ)।

Q40. Which of the following statements is NOT true about microwaves?

Microwaves के बारे में निम्न में से कौन-सा कथन सही नहीं है?

(A) They are electromagnetic waves with frequencies from 300 GHz to 400 THz **(B)** Propagation is line-of-sight

(C) Very high-frequency waves cannot penetrate walls **(D)** Use of certain portions of the band requires permission from authorities

[Source: Repo: CN Topic Test-04 (UNIT TEST-4) — options cleaned from source]

✔ **Correct: (A) They are electromagnetic waves with frequencies from 300 GHz to 400 THz** — Microwaves range roughly 300 MHz to 300 GHz — '300 GHz to 400 THz' is the infrared region, so this statement is false.

Microwaves की range लगभग 300 MHz से 300 GHz है — '300 GHz से 400 THz' infrared क्षेत्र है, इसलिए यह कथन गलत है।

★ **Nukta:** Microwaves: 300 MHz-300 GHz, line-of-sight, no wall penetration, licensed bands. Exam asks the odd range out.

Microwaves: 300 MHz-300 GHz, line-of-sight, दीवार नहीं भेदते, licensed bands। Exam गलत range पूछता है।

Q41. Which transmission medium is used in Bluetooth technology?

Bluetooth technology में कौन-सा transmission medium उपयोग होता है?

(A) Radio links **(B)** Microwave links

(C) VSAT communication **(D)** Fiber optic

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)]

✔ **Correct: (A) Radio links** — Bluetooth (IEEE 802.15.1) uses short-range radio links in the 2.4 GHz ISM band.

Bluetooth (IEEE 802.15.1) 2.4 GHz ISM band में short-range radio links उपयोग करता है।

★ **Nukta:** Bluetooth = 2.4 GHz radio, WPAN (802.15), ~10-100 m range, frequency hopping.

Bluetooth = 2.4 GHz radio, WPAN (802.15), ~10-100 m range, frequency hopping।

Q42. Which physical-layer technique allows multiple users to share the same bandwidth simultaneously?

कौन-सी physical-layer technique कई users को एक ही bandwidth साथ-साथ share करने देती है?

(A) TDMA **(B)** FDMA

(C) CDMA **(D)** ARQ

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)]

✔ **Correct: (C) CDMA** — In CDMA every user transmits over the SAME bandwidth at the SAME time, separated by unique codes.

CDMA में हर user एक ही bandwidth पर एक ही समय में transmit करता है, अलग-अलग codes से अलग किया जाता है।

★ **Nukta:** Sharing methods: FDMA = frequency bands, TDMA = time slots, CDMA = codes (same band+time).

Sharing methods: FDMA = frequency bands, TDMA = time slots, CDMA = codes (same band+time)।

Q43. Select the FALSE statement about CDMA technology: I. CDMA optimises the use of available bandwidth as it transmits over the entire frequency range. II. Each user in a CDMA system uses a different code to modulate their signal. III. CDMA offers less security compared to GSM technology. IV. A SIM card is always required for a CDMA device to work.

CDMA technology के बारे में गलत कथन चुनिए: I. CDMA उपलब्ध bandwidth के उपयोग को अनुकूलित करता है क्योंकि यह पूरी frequency range पर transmit करता है। II. CDMA system में हर user अपने signal को modulate करने के लिए अलग code उपयोग करता है। III. CDMA, GSM की तुलना में कम security देता है। IV. CDMA device के काम करने के लिए SIM card हमेशा आवश्यक है।

(A) Only (I) **(B)** Only (II)

(C) (I) and (II) **(D)** (III) and (IV)

[Source: Random Test- Rajasthan PYQ (OCR), p11 — Q83 (statements recovered from screenshot)]

✔ **Correct: (D) (III) and (IV)** — Statements III and IV are false: CDMA offers STRONGER security than GSM (spread spectrum + codes), and CDMA devices can work without a SIM (older CDMA handsets).

कथन III और IV गलत हैं: CDMA, GSM से मज़बूत security देता है (spread spectrum + codes), और CDMA devices बिना SIM के भी चल सकते हैं (पुराने CDMA handsets)।

★ **Nukta:** Rajasthan PYQ verified key (D). True: I (spread over full band) and II (unique codes per user). False: III (CDMA more secure) and IV (SIM not mandatory).

Rajasthan PYQ verified key (D)। सही: I (पूरे band पर spread) और II (हर user का unique code)। गलत: III (CDMA अधिक सुरक्षित) और IV (SIM अनिवार्य नहीं)।

★ Nukte Rapid-Fire — Exam Pointers (quick revision)

One-line question (left) with its exam answer + trick (right). Revise these before the exam.

Attenuation / distortion / noise? Attenuation / distortion / noise?	✔ Strength loss / shape change / unwanted signals. Noise can only be MINIMISED, never removed. Strength loss / shape change / unwanted signals। Noise केवल कम होती है, कभी पूरी तरह नहीं।
Crosstalk cause? Crosstalk का कारण?	✔ Electromagnetic coupling between nearby pairs — minimised by tighter twists. पास की pairs के बीच electromagnetic coupling — tighter twists से कम होता है।
Manchester encoding? Manchester encoding?	✔ Mid-bit transition = self-clocking + DC-free + noise-resistant (Physical layer). Mid-bit transition = self-clocking + DC-free + noise-resistant (Physical layer)।
Encoding redundancy order? Encoding redundancy क्रम?	✔ 1B2B (100%) > 4B5B/8B10B (25%) > 5B6B (20%). Gigabit fiber = 8B/10B. 1B2B (100%) > 4B5B/8B10B (25%) > 5B6B (20%)। Gigabit fiber = 8B/10B।
Scrambler use? Scrambler का उपयोग?	✔ Make the line signal look random (break long runs) — helps clock recovery. Not compression. Line signal को random बनाना (लंबे runs तोड़ना) — clock recovery में मदद। Compression नहीं।
Shannon & Nyquist formulas? Shannon और Nyquist सूत्र?	✔ $C = B \cdot \log_2(1+SNR)$ (noisy); $C = 2B \cdot \log_2(L)$ (noiseless). $C = B \cdot \log_2(1+SNR)$ (noisy); $C = 2B \cdot \log_2(L)$ (noiseless)।
Bit rate vs baud rate? Bit rate बनाम baud rate?	✔ Bit rate = baud × bits per signal unit. Bit rate = baud × प्रति signal unit bits।
Time to send n bits at b bps? b bps पर n bits का समय?	✔ t = n/b seconds. t = n/b seconds।

Analog signal parameters? Analog signal parameters?	✔ Amplitude (height), Frequency (cycles/sec, Hz), Phase (start), Wavelength (distance/cycle). Amplitude (ऊँचाई), Frequency (cycles/sec, Hz), Phase (शुरुआत), Wavelength (दूरी/cycle)।
Quantization noise in? Quantization noise किसमें?	✔ PCM (Pulse Code Modulation) — rounding analog samples to levels. PCM (Pulse Code Modulation) — analog samples को levels पर गोल करना।
Baseband vs broadband? Baseband बनाम broadband?	✔ Baseband = digital signals directly. Broadband = modulated analog carriers (FDM). Baseband = digital signals सीधे। Broadband = modulated analog carriers (FDM)।
Bandwidth vs throughput? Bandwidth बनाम throughput?	✔ Bandwidth = max capacity; Throughput = actual data per second (exam often mixes them). Bandwidth = अधिकतम capacity; Throughput = वास्तविक data प्रति सेकंद (exam में अक्सर मिलाया जाता है)।
Fiber's 3 superpowers? Fiber की 3 शक्तियाँ?	✔ EMI-immune, tap-proof, highest bandwidth (via total internal reflection). EMI-मुक्त, tap-सुरक्षित, सर्वोच्च bandwidth (total internal reflection से)।
TIR condition? TIR शर्त?	✔ Angle of incidence > critical angle (denser to rarer medium). Angle of incidence > critical angle (denser से rarer medium)।
UTP / coax / fiber distance limits? UTP / coax / fiber दूरी सीमाएँ?	✔ 100 m / 185-500 m / up to tens of km (single-mode). 100 m / 185-500 m / दसियों km तक (single-mode)।
Connectors? Connectors?	✔ RJ-45 (twisted pair), BNC (coax), SC/ST/LC (fiber), RJ-11 (telephone). RJ-45 (twisted pair), BNC (coax), SC/ST/LC (fiber), RJ-11 (telephone)।

Coax reflection cause? Coax reflection का कारण?	✔ Wrong termination resistance (impedance mismatch). गलत termination resistance (impedance mismatch)।
Single-mode vs multi-mode fiber? Single-mode बनाम multi-mode fiber?	✔ Single = long haul (km, backbone). Multi = short LAN distances (modal dispersion). Single = लंबी दूरी (km, backbone)। Multi = छोटी LAN दूरी (modal dispersion)।
Wi-Fi rate order? Wi-Fi rate क्रम?	✔ b (11 Mbps) < g (54) < n (600) < ac (multi-Gbps). b (11 Mbps) < g (54) < n (600) < ac (multi-Gbps)।
Line-of-sight wireless? Line-of-sight wireless?	✔ Infrared (and microwave links) need LOS. Radio (Wi-Fi/Bluetooth) passes walls. Infrared (और microwave links) को LOS चाहिए। Radio (Wi-Fi/Bluetooth) दीवारों से गुजरता है।
Microwave band? Microwave band?	✔ 300 MHz - 300 GHz (NOT 300 GHz-400 THz — that is infrared!). 300 MHz - 300 GHz (300 GHz-400 THz नहीं — वह infrared है!)।
Bluetooth = ? Bluetooth = ?	✔ 2.4 GHz short-range radio (IEEE 802.15.1 WPAN). 2.4 GHz short-range radio (IEEE 802.15.1 WPAN)।
FDMA / TDMA / CDMA? FDMA / TDMA / CDMA?	✔ Frequency bands / time slots / unique codes (same band & time). Frequency bands / time slots / unique codes (same band और time)।